

CS4800: Designing Sustainable ICT Systems

Lecture 7.2: Sustainable Data Storage, Part 2

October 16, 2026

Lecturer: Przemysław Pawełczak

Authors: Alex Nedelcu, Przemysław Pawełczak

CS4800-EWI@tudelft.nl

What will you learn in this lecture?

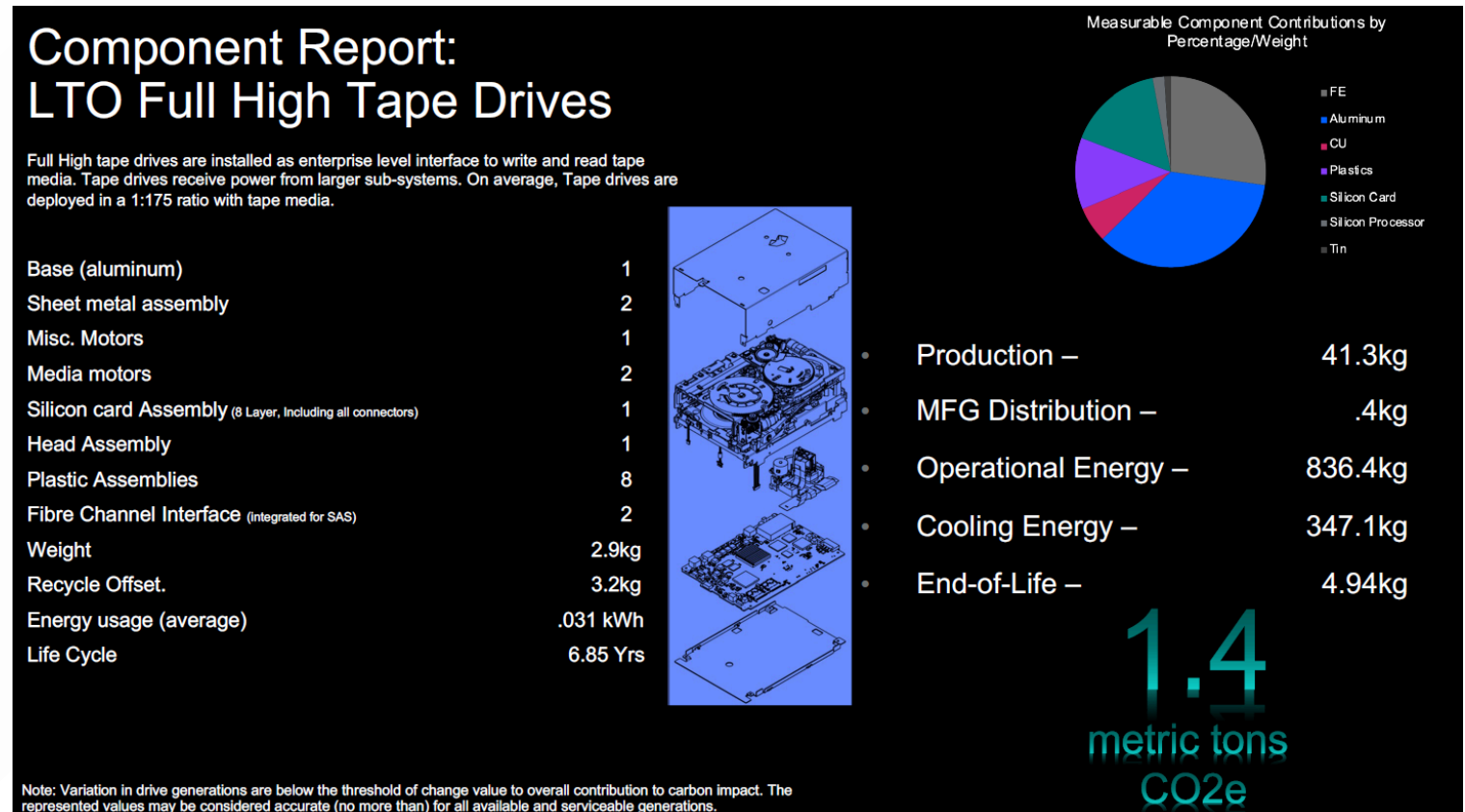
Lecture 7.2: Sustainable Data Storage, Part 2

- Environmental impacts: carbon footprint, HDD/SSD trade-offs, critical raw materials
- Industry trends: storage demand, areal density
- Prospective solutions: lifetime extension, data storage strategy, novel technologies

What are the impacts of data storage?

Data storage impacts: overview

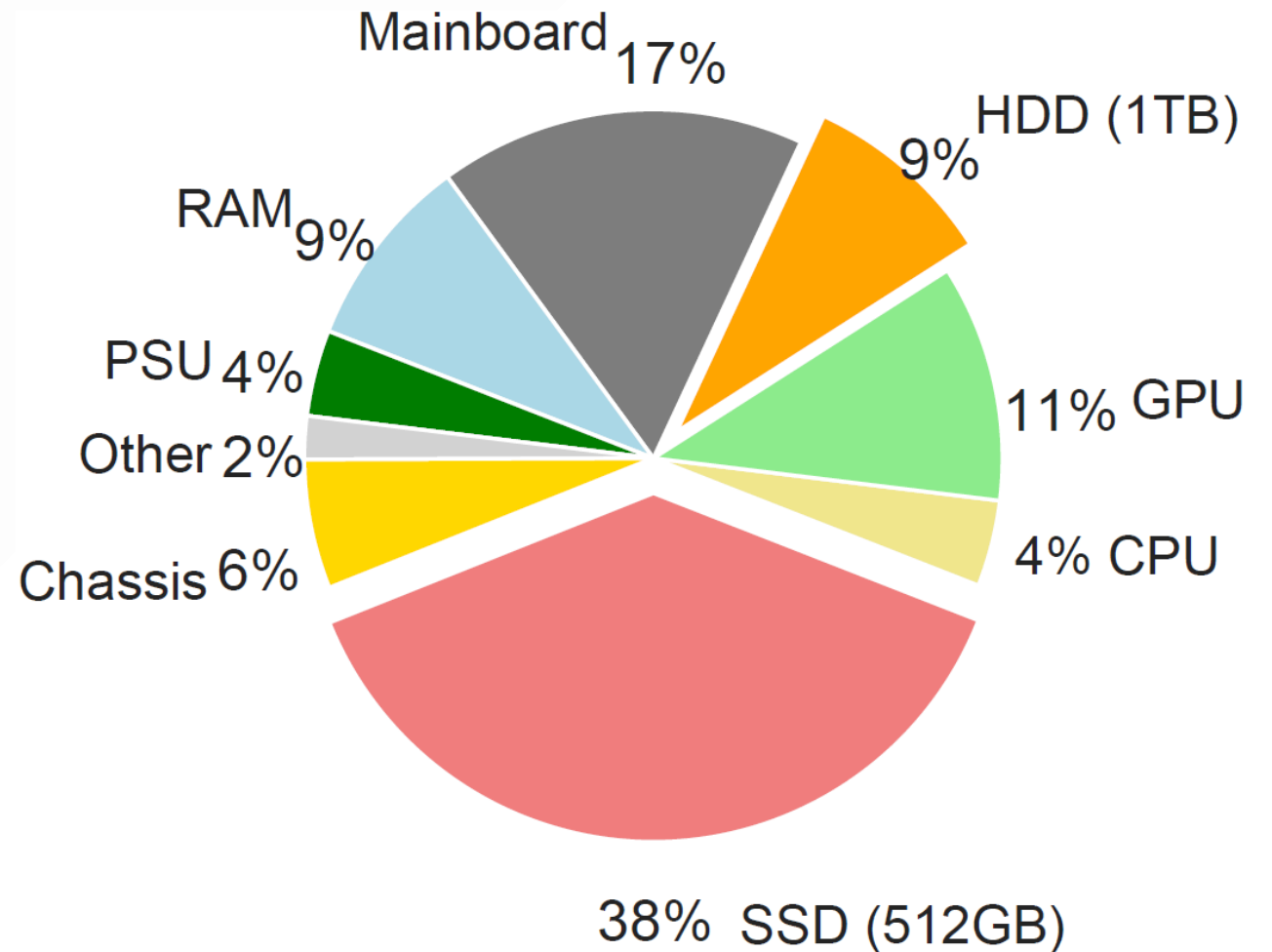
- Data storage technologies are physical devices with environmental impacts:
 - Carbon footprint: embedded and operational emissions
 - Material footprint: critical raw materials
 - Other: water footprint etc



Example: environmental impacts related to an LTO tape drive

Data storage impacts: carbon footprint

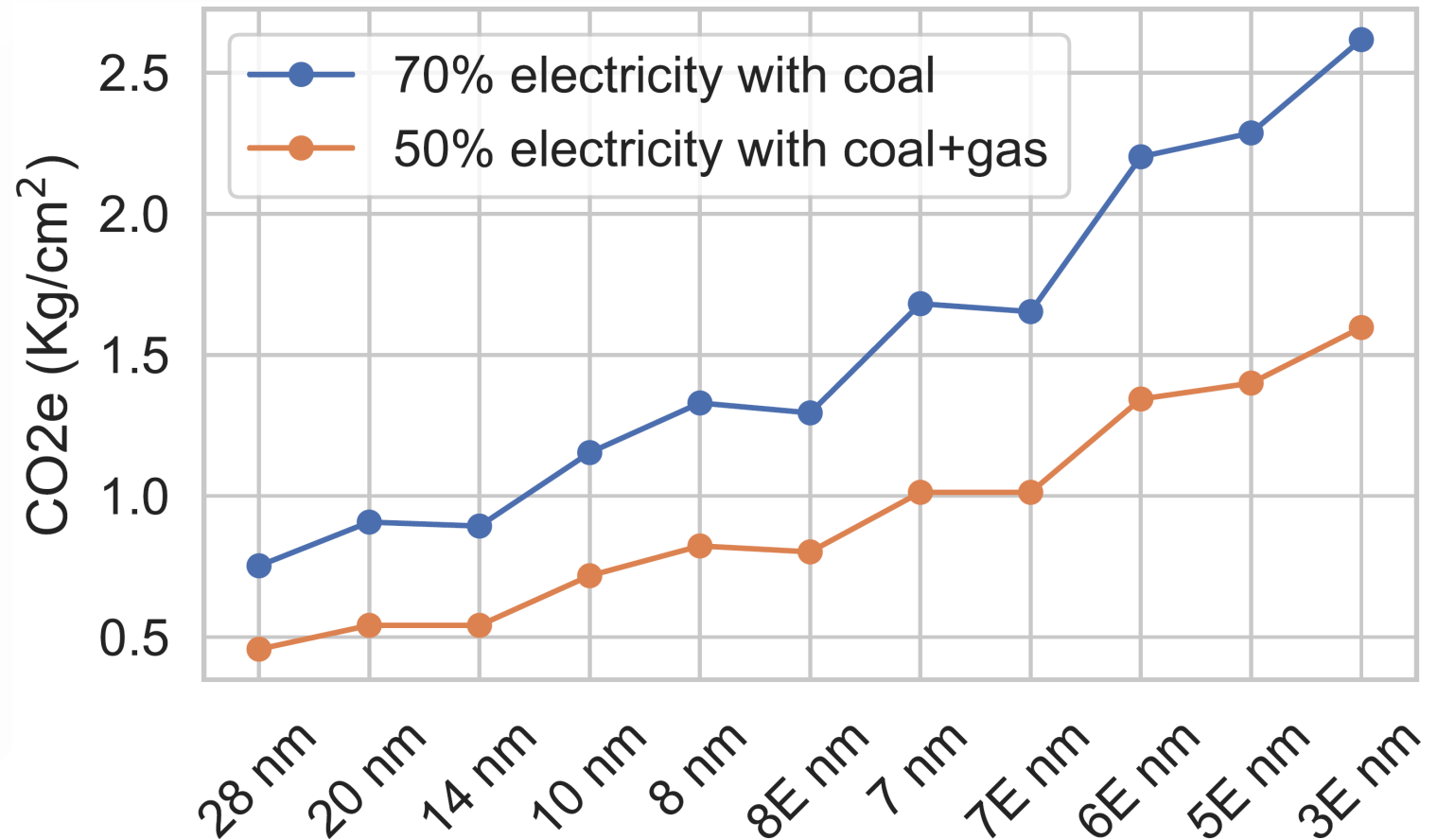
- Data storage is important contributor to both embedded and operational emissions for ICT systems:
 - User devices: HDD+SSD cause [~50% of embedded emissions](#)
 - Microsoft Azure servers: 33% of operational, 61% of embedded emissions are [storage-related](#)



[Breakdown of desktop computer embedded emissions](#)

Data storage impacts: embedded emissions

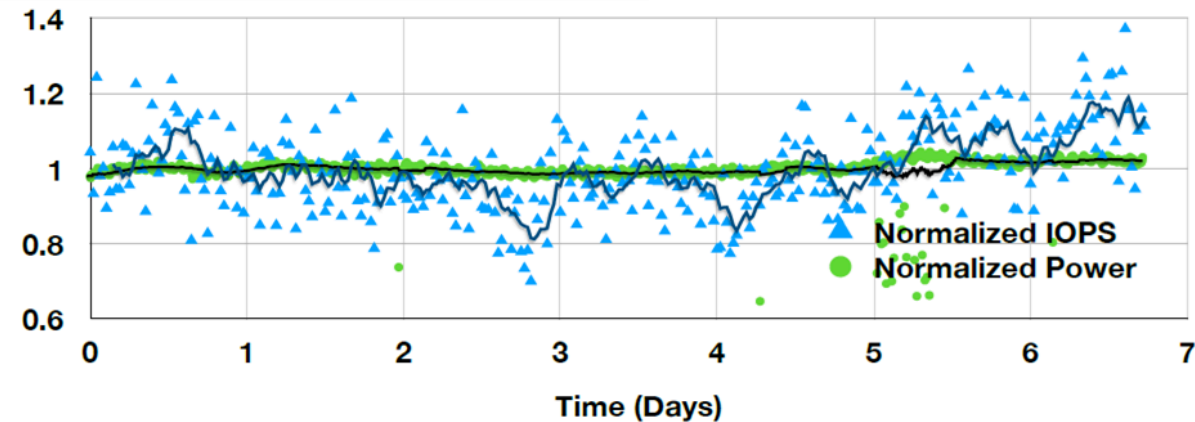
- Material sourcing, processing, and device manufacturing dominates SSD lifecycle emissions
- Semiconductor fabrication: transistor scaling, 3D die stacking
 - High-precision, energy-intensive processes!
 - Emissions increase with capacity despite constant form factor



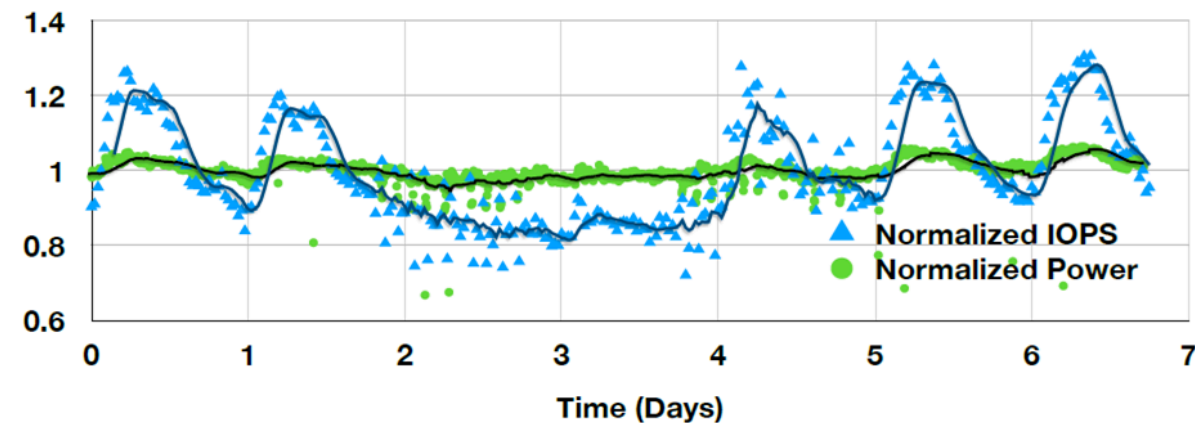
Carbon intensity of semiconductor memory increases as transistors become smaller

Data storage impacts: operational emissions

- All storage technologies consume energy to function → grid electricity has carbon intensity
- Storage power consumption is constant:
 - HDDs continue spinning regardless of IO operations
 - SSDs somewhat more power proportional



(a) HDD cluster

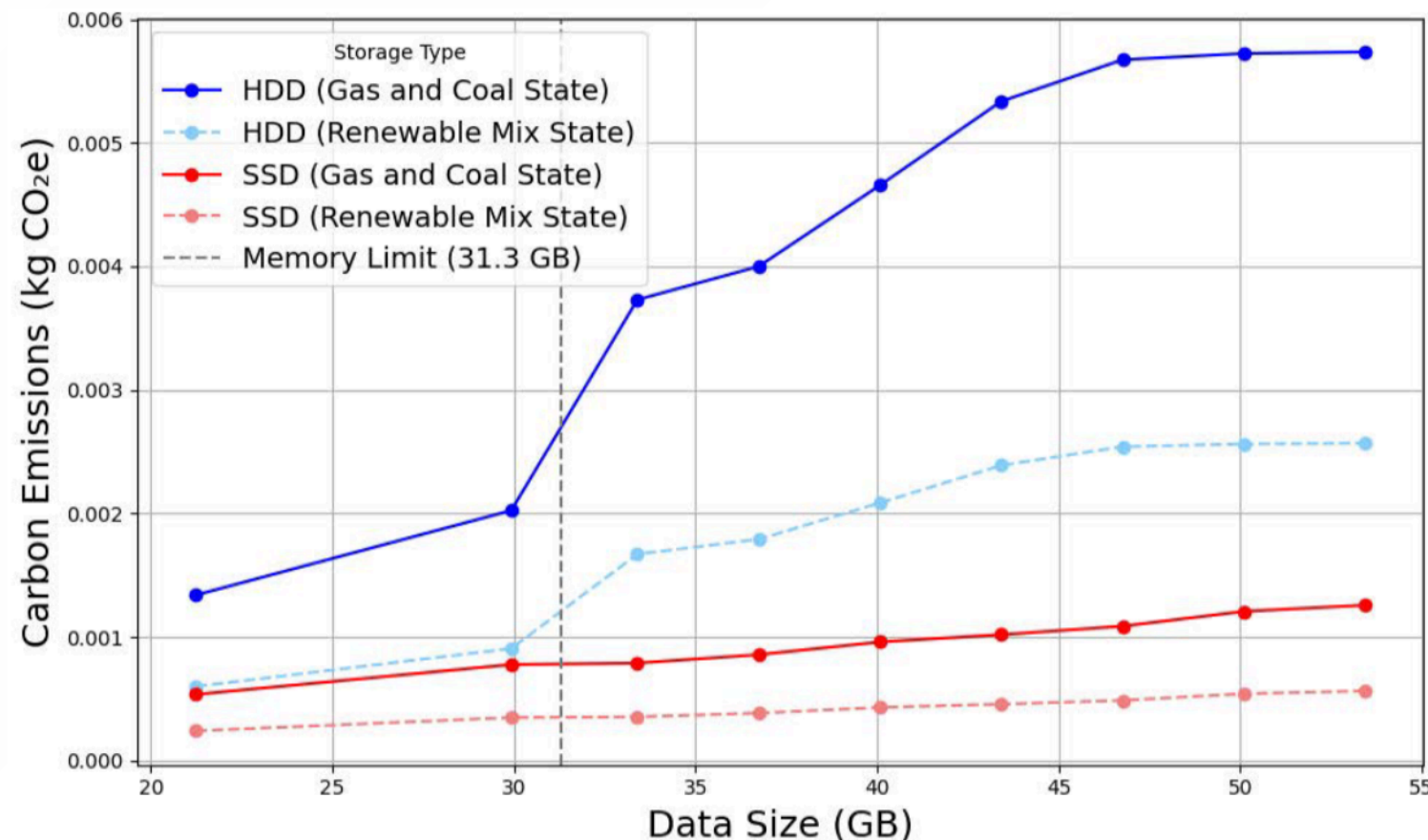


(b) SSD cluster

Normalized power draw and IOPS for Microsoft
Azure HDD and SSD clusters

Data storage impacts: HDD vs SSD emissions

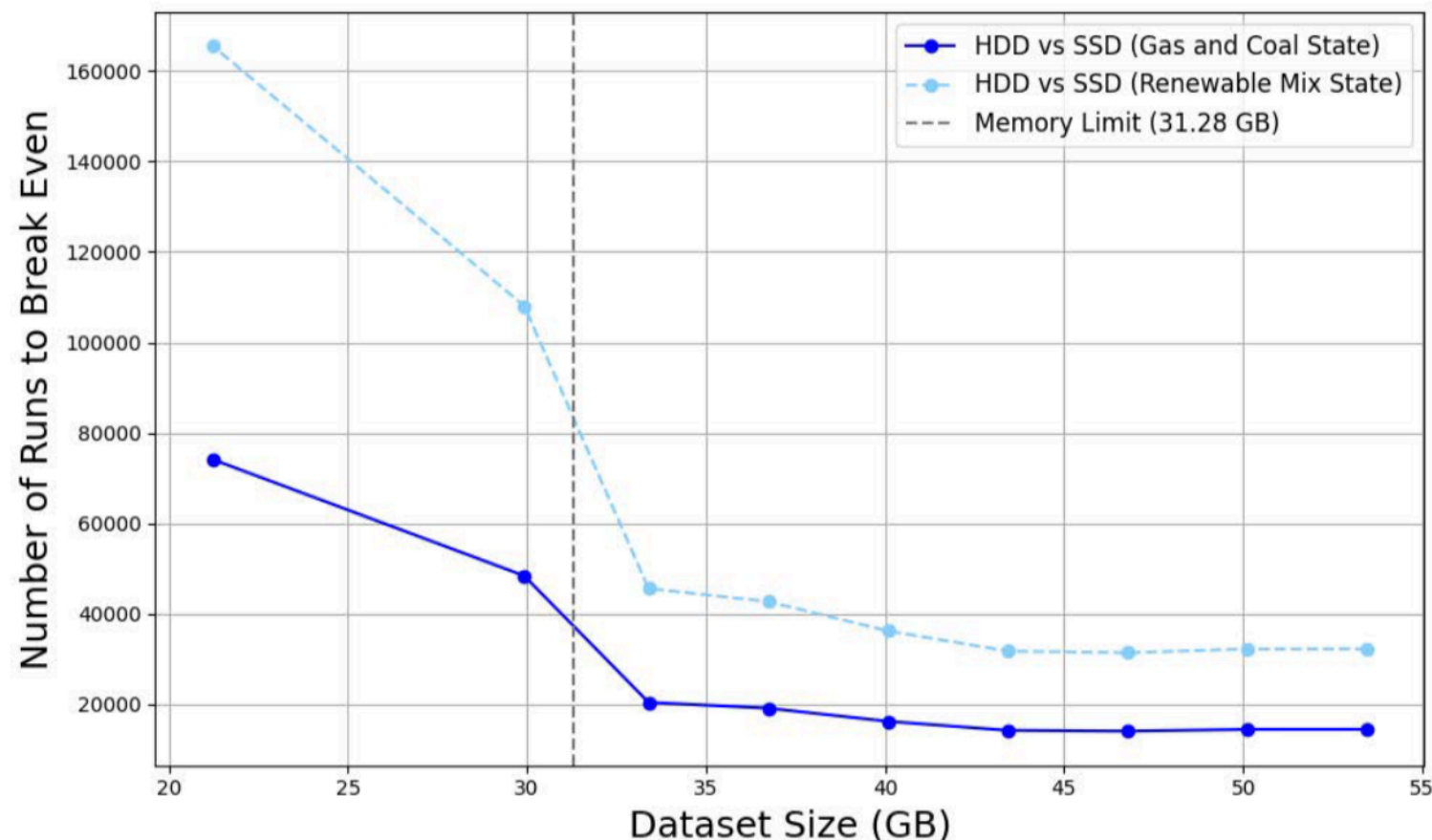
- Complex trade-offs between HDDs and SSDs:
 - Embedded emissions: SSDs always higher due to complex fabrication processes
 - Operational emissions: HDDs typically have higher power draws to maintain spin rates
 - Bandwidth limitations may lead to multiple HDDs being required for same workload!



HDD operational emissions during machine learning runs increase drastically after hitting memory limit

Data storage impacts: HDD vs SSD emissions

- Optimal technology over lifecycle: depends on many factors
 - Grid carbon intensity during manufacturing
 - Grid carbon intensity during operations
 - Operation types
 - Lifetime/number of cycles

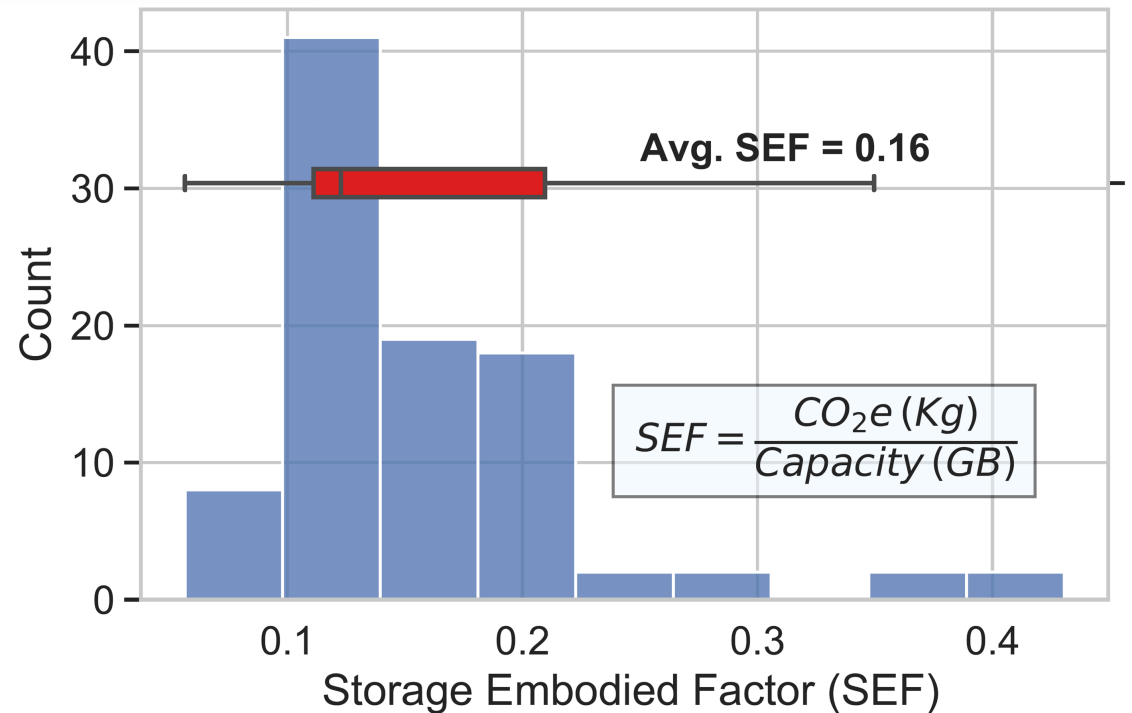


Number of ML runs required for SSD emissions to break even with HDD emissions depends on grid intensity, workload size

Exercise: HDD vs SSD emissions

After how many runs do equivalent capacity SSDs break even with HDDs?

- SSD: storage embodied factor (SEF) of 0.15 kgCO₂/GB, constant power draw of 5 W, runtime of 30 minutes
- HDD: SEF of 0.01 kgCO₂/GB, constant power draw of 4 W, runtime of 2 hours
- Grid carbon intensity: 0.3 kgCO₂/kWh (fossil-heavy) or 0.15 kgCO₂/kWh



[Average storage embodied factor for SSDs](#)

Data storage impacts: tape emissions

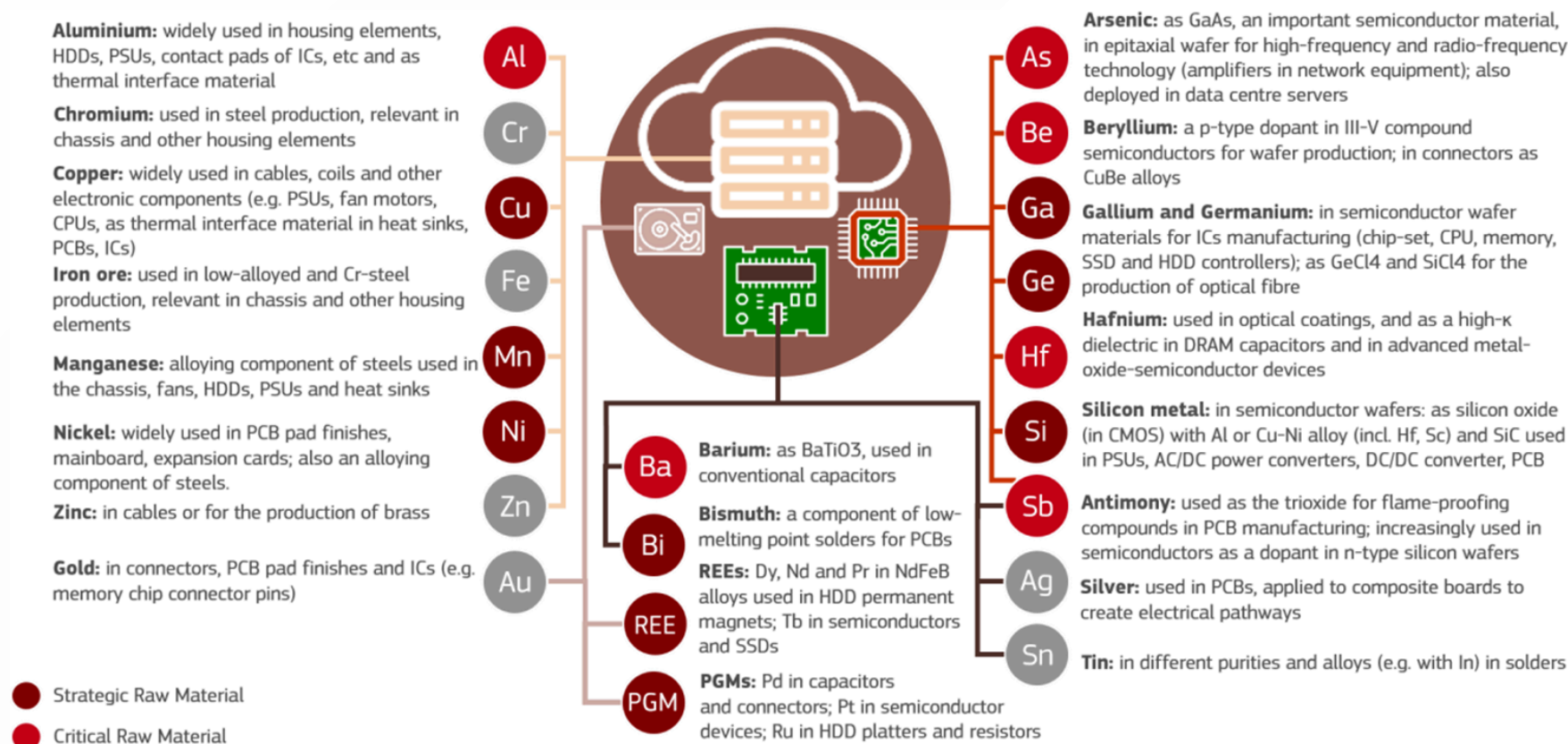
- Magnetic tape: [relatively low embedded emissions](#), only consumes power when active
 - Embedded carbon per TB per year: HDD < tape < SSD
 - Operational carbon per TB per year: tape < SSD < HDD
- ... but tape system includes more than just cartridges!
- ... but different use-cases and requirements (e.g. [latency](#)) for each technology!

Storage media ⁷	Operating power in watts	Power efficiency (W/TB)
SSD	20	0.5
Hard drive	9.6	0.32
LTO (Tape)	37	1.1

[Device operating power and average power consumed to read/write 1 TB of data](#)

Data storage impacts: critical raw materials

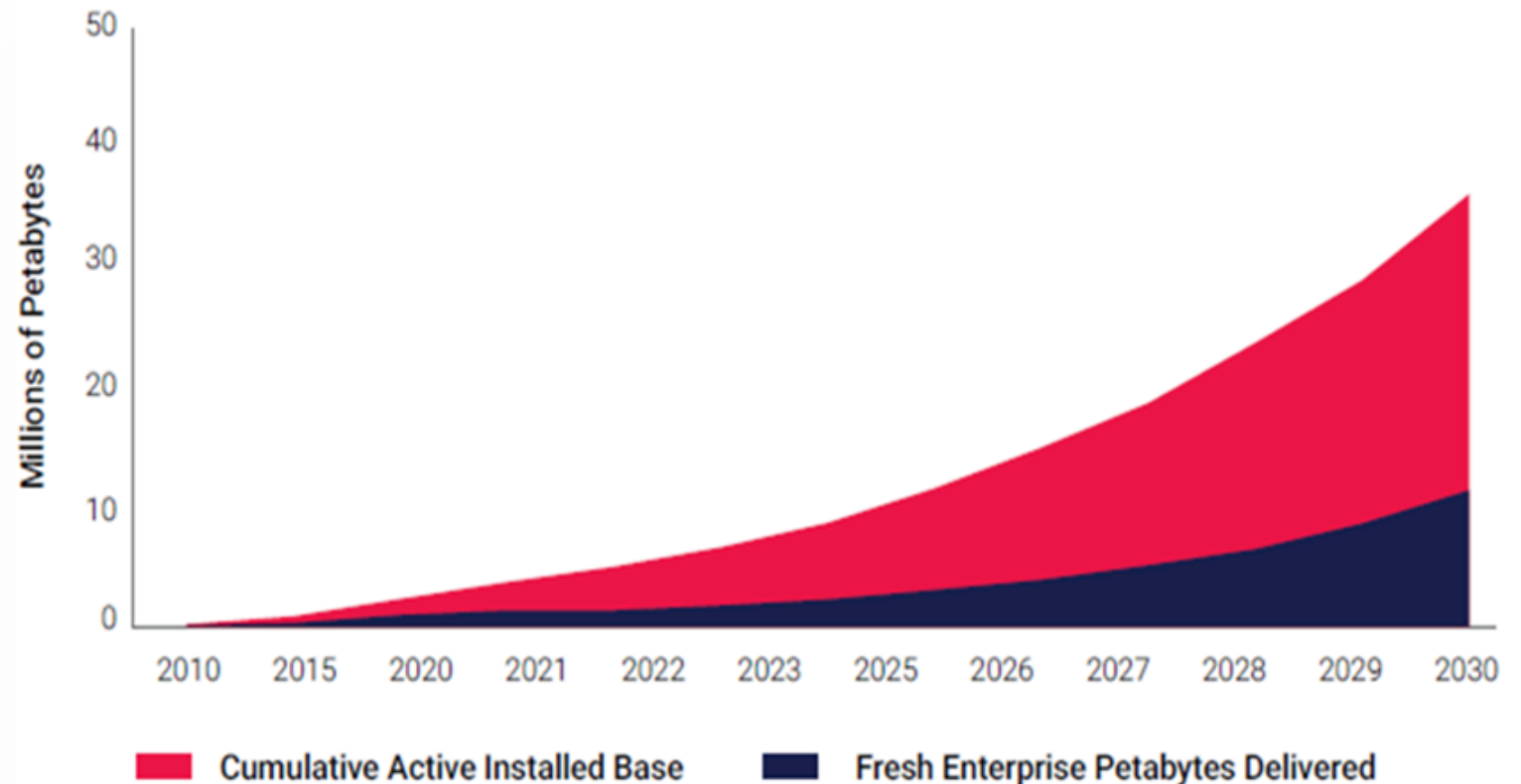
- Silicon [dominant component in SSDs](#), rare earth metals used in HDD magnets
- Low demand relative to global production, but geopolitical supply risks remain



What are the most important trends in the data storage industry?

Industry trends: storage demand

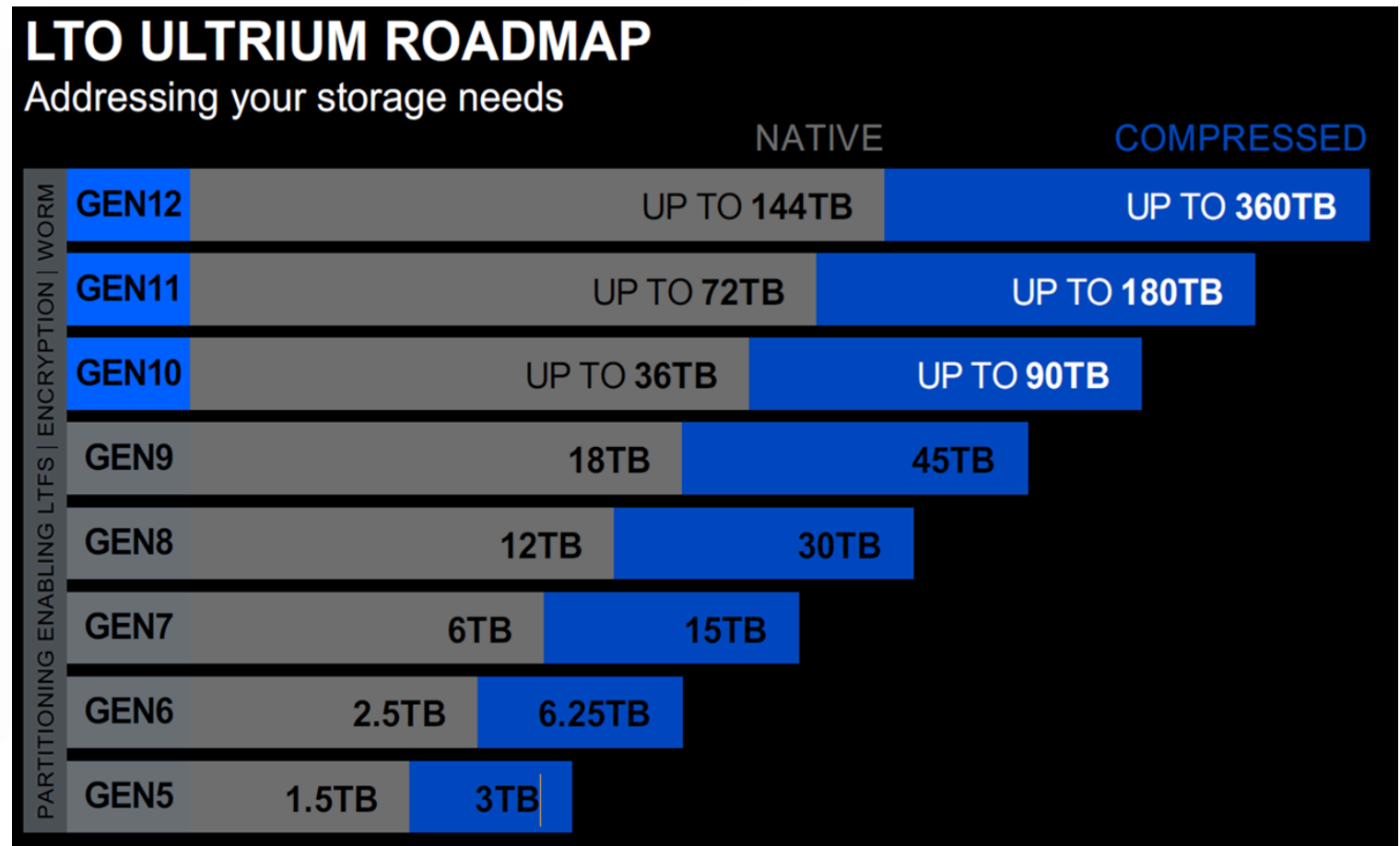
- [All projections point](#) to continuous growth in demand for storage
- Growth across all use-cases and technologies
 - SSDs remain dominant, HDD and tape still used depending on use-case



[Projection of new and cumulative storage capacity until 2030](#)

Industry trends: magnetic tape

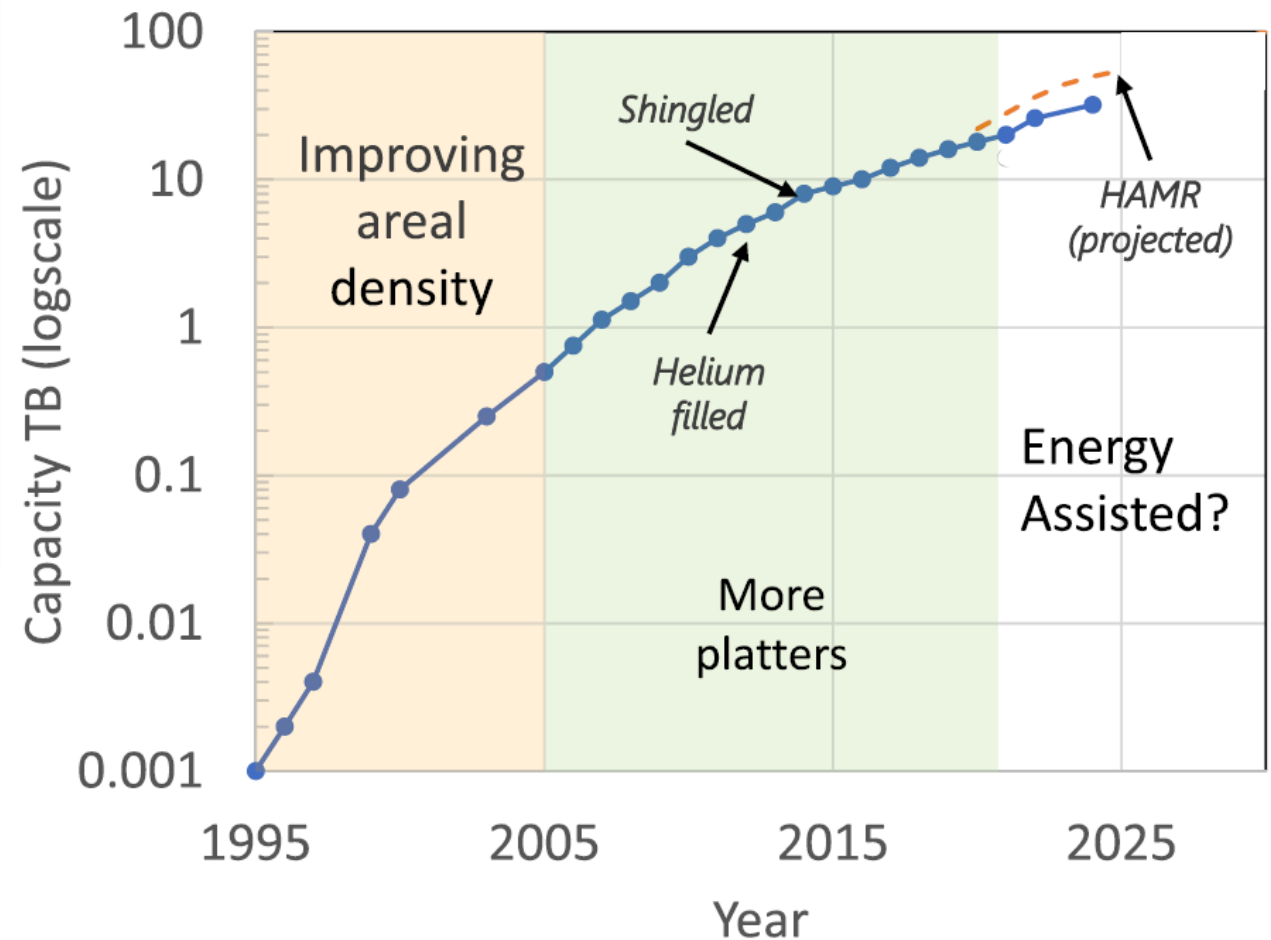
- Fundamentals expected to remain the same
- LTO-10 → LTO-14: increasing areal and volume density in same form factor
- Volume density doubles with each generation, but manufacturing complexity also increases



Magnetic tape capacity has continued to increase over time

Industry trends: HDDs

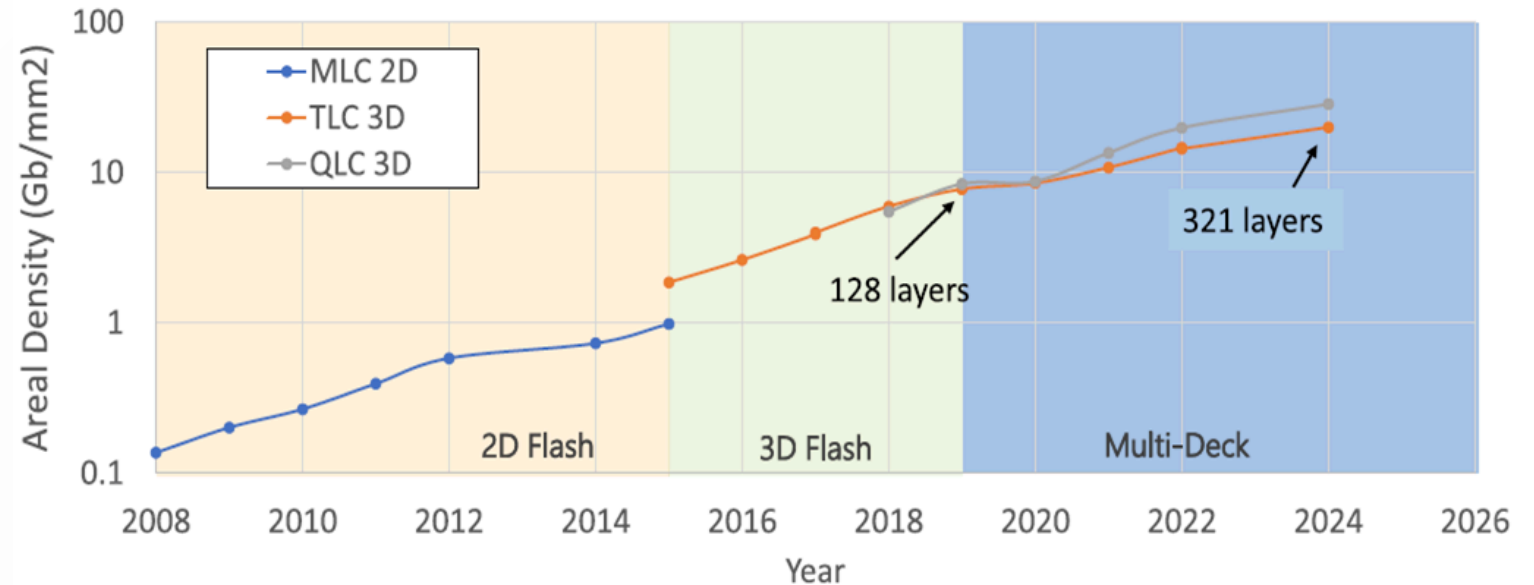
- Increasing areal density and volume density in same form factor
 - HAMR: using [energy input](#) (heat) to support higher densities
- Access rates (IOPS) [do not improve at same rates as capacity](#)
→ high-bandwidth workloads do not benefit!



[HDD capacity has continued to increase over time](#)

Industry trends: SSDs

- [Diminishing returns](#) to increasing areal density:
 - Historical gains due to 2D → 3D stacking, increased number of bits per cell
 - Higher density increases manufacturing complexity and [vastly reduces](#) lifetime!

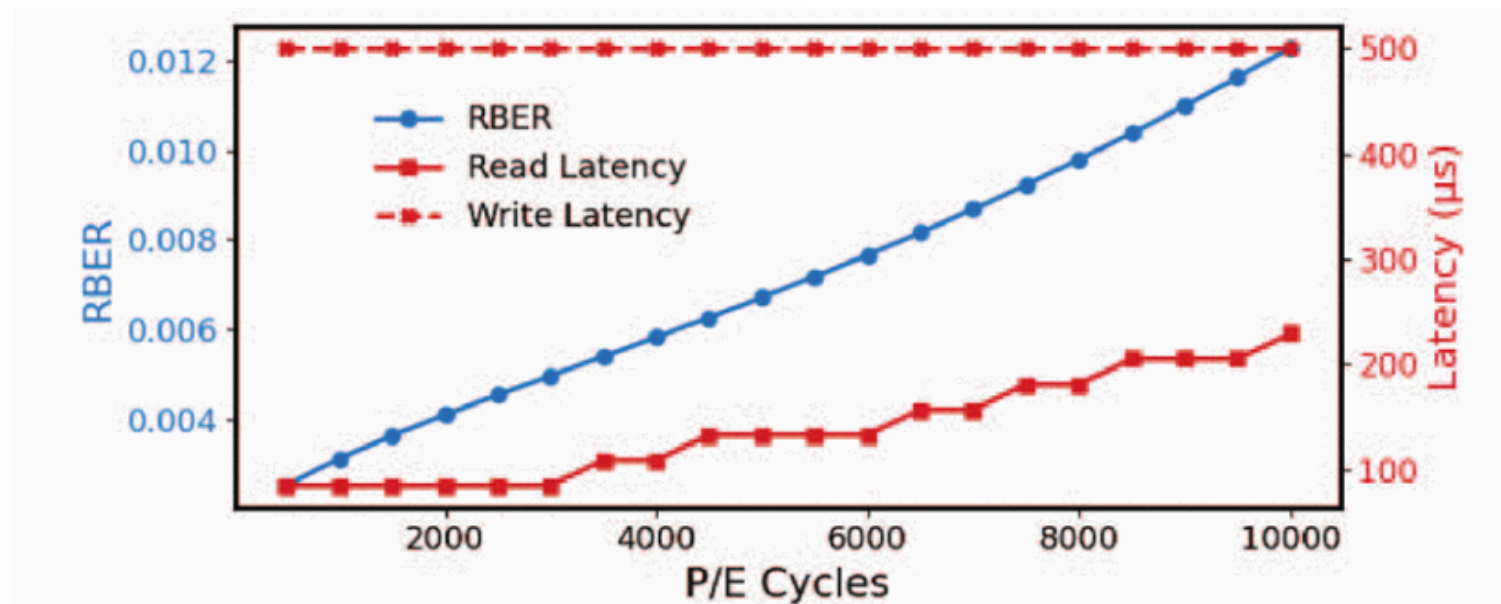


[Memory cell areal density increases and linked innovations](#)

What solutions do we have at our disposal?

Prospective solutions: lifetime extension

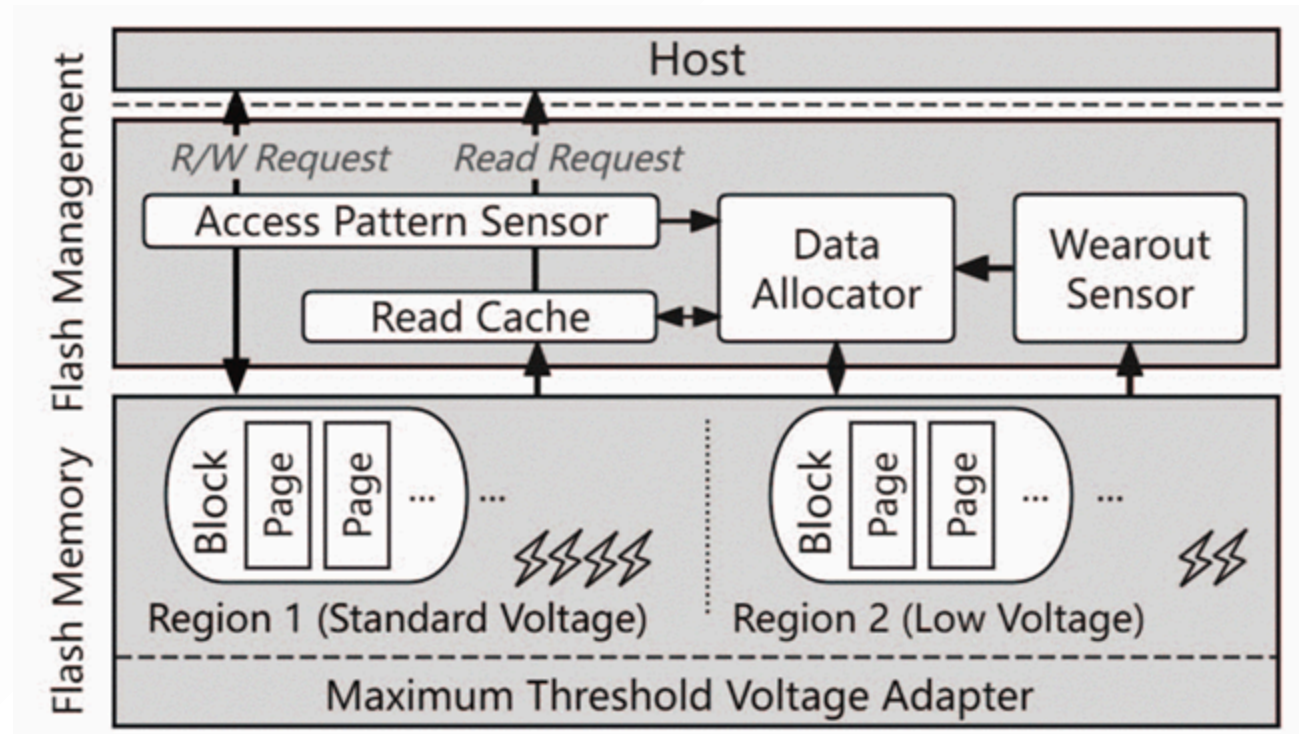
- SSDs: failure rates [increase](#) with number of read/write cycles
- HDDs: failure rates [unknown](#), typically decommissioned before failure!
- Doubling SSD lifetime doubles write operations → limited potential
- Still possible for fault-tolerant design to [accept partial failure](#), allow individual HDD sectors/SSD cells to grow defects



[Raw bit error rate \(RBER\) and latency increase with each program/erase cycle](#)

Prospective solutions: data allocation

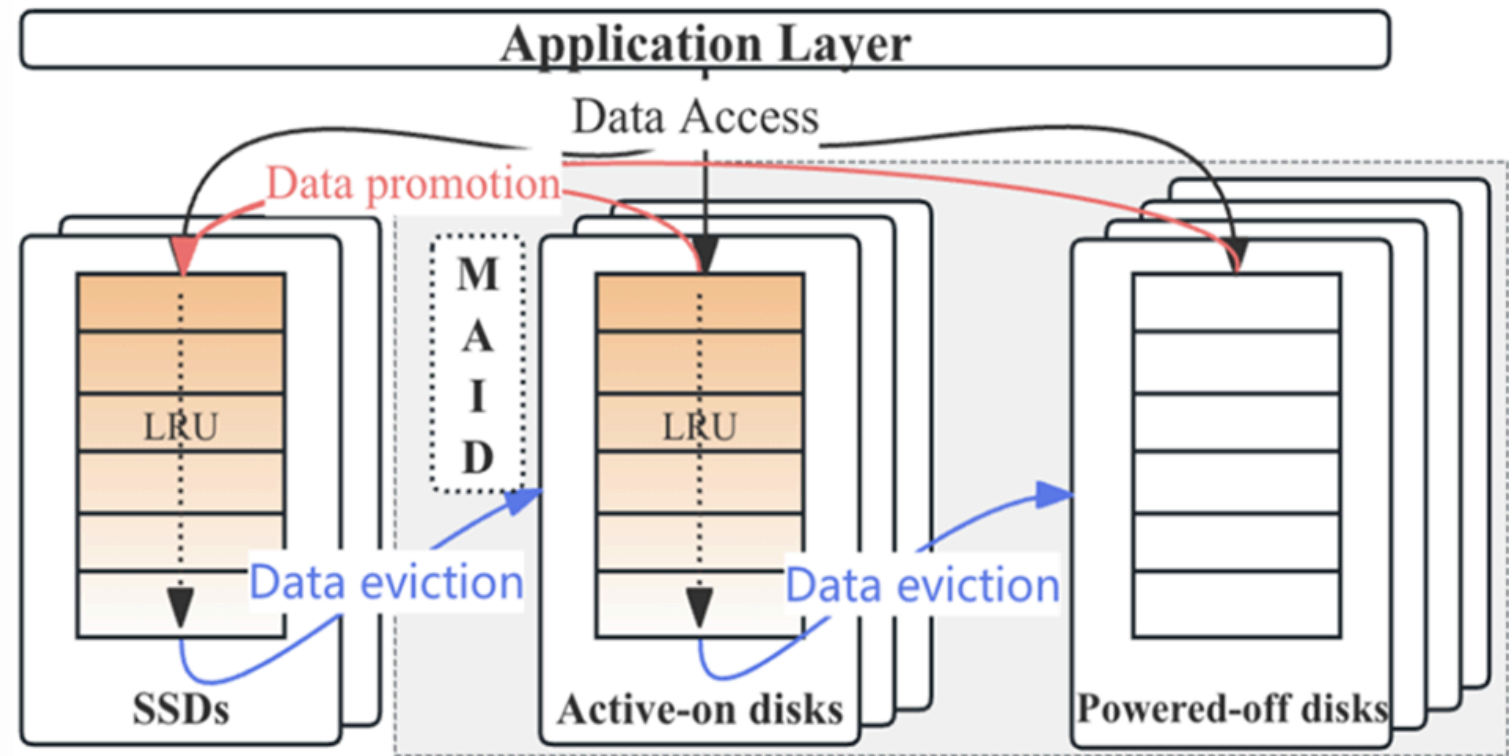
- Higher threshold voltage in SSD read/write operations [reduces latency and noise](#)
 - ...but also causes more defects!
- Solution: divide disk into two regions, [dynamically migrate](#) data to low-voltage partition based on observed access patterns
- [Wear leveling](#) between regions extends lifetime



[Architecture for allocating stored data depending on component wearout](#)

Prospective solutions: data allocation

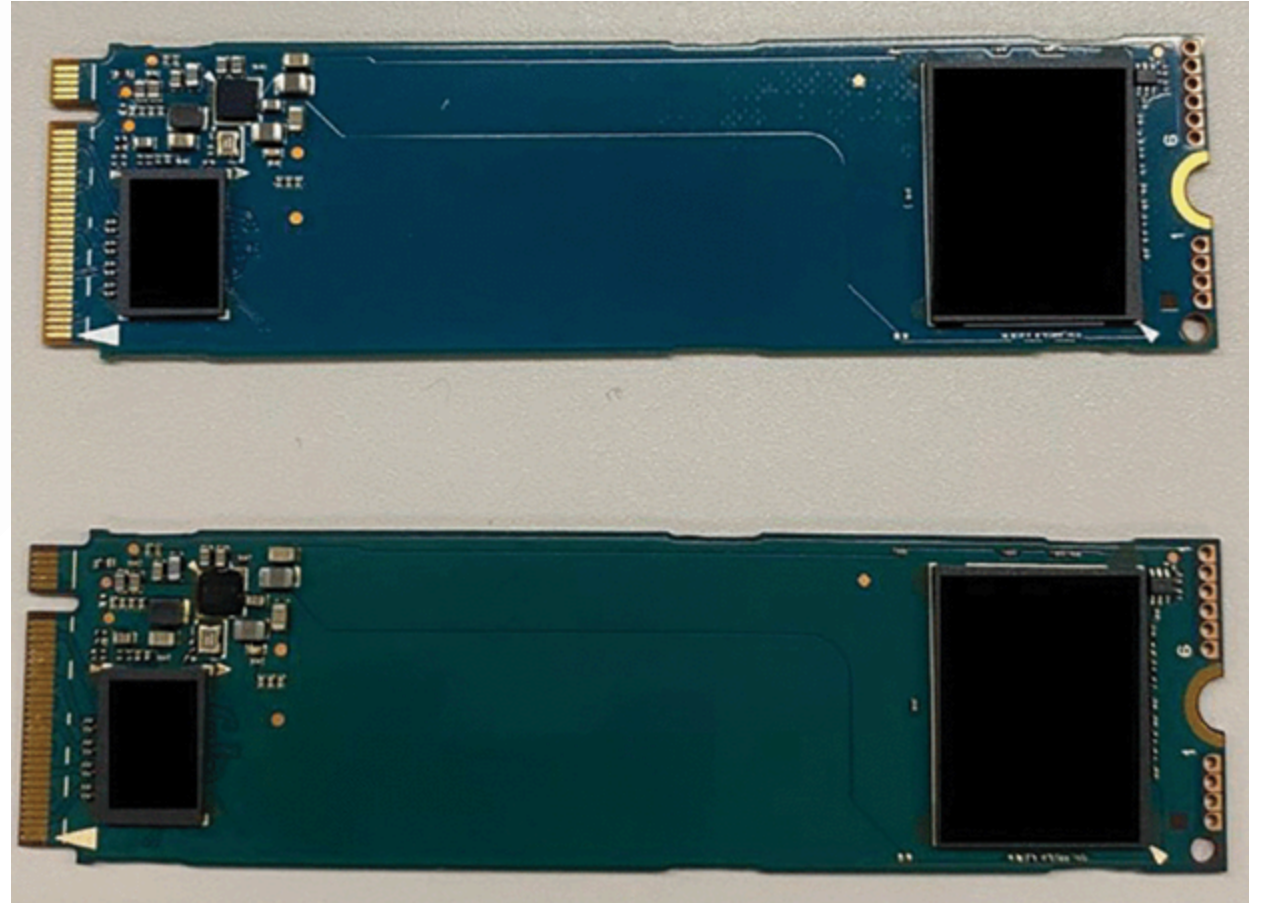
- [Massive arrays of idle disks](#) (MAID): three-tiered architecture
 - Hot data: SSD
 - Warm data: active HDD
 - Cold data: idling HDD
- Lower power draw, but higher latency if trying to access data in idle HDDs
- ... but: in [service-level agreement](#) (SLA), user purchases performance (latency) guarantee!



Data is constantly promoted to or evicted from active storage

Prospective solutions: SSD annealing

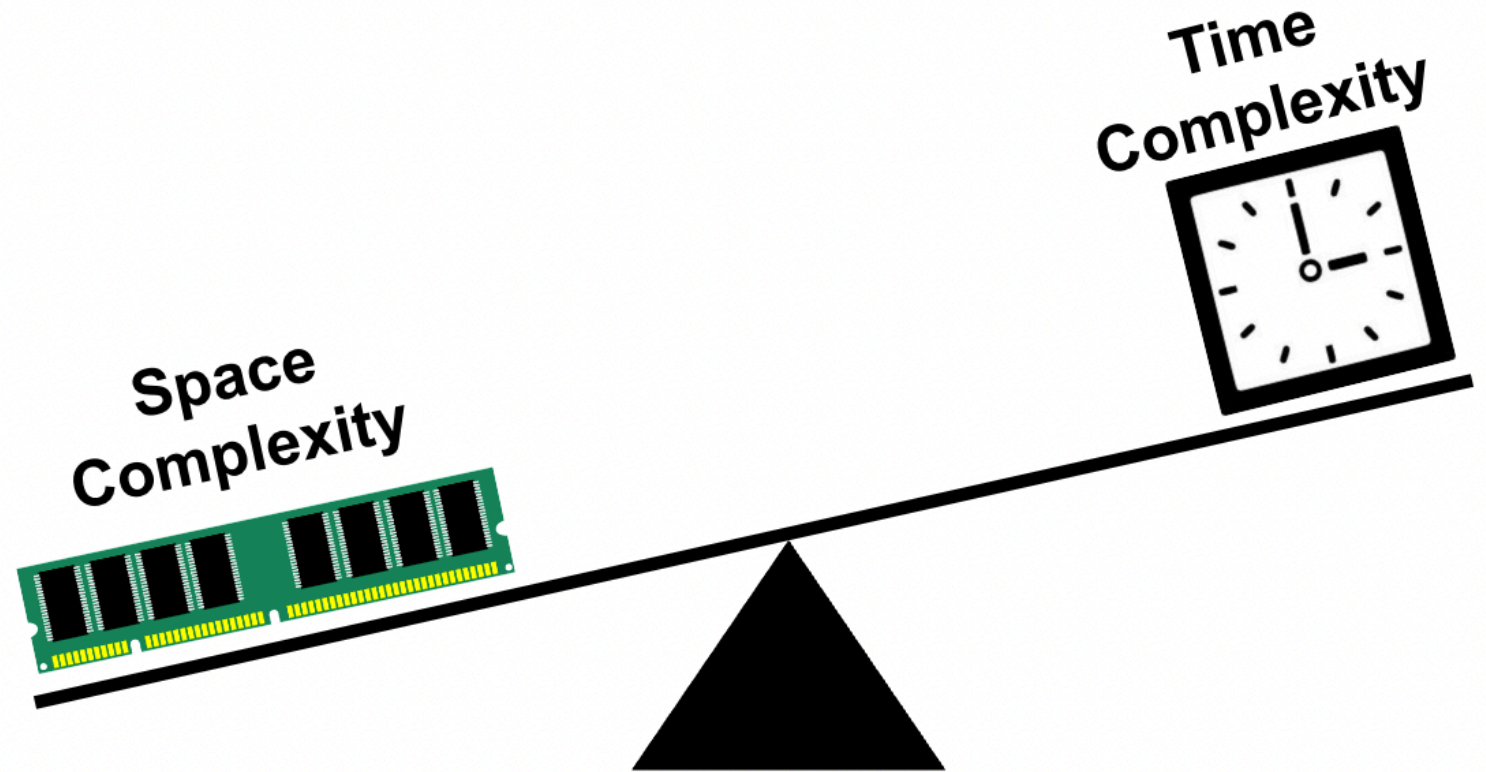
- SSD [annealing](#): bake the SSD to eliminate defects accumulated during read/write operations
 - How many annealing cycles can the PCB admit?
- Effectively recovers performance even at low (150°C) temperatures!
- Requires manual processing



It may look crispy, but it gets the job done!

Prospective solutions: space/time trade-off

- Space-time trade-off
[principle](#): 'one can often achieve a reduction in time if one is willing to sacrifice space or vice versa'
- Data compression: modest CPU compute load, [enormous storage savings](#)
- Reproducing numerical experiments might consume [less energy](#) than long-term storage



[High emissions involved with data storage might motivate users to minimize space complexity](#)

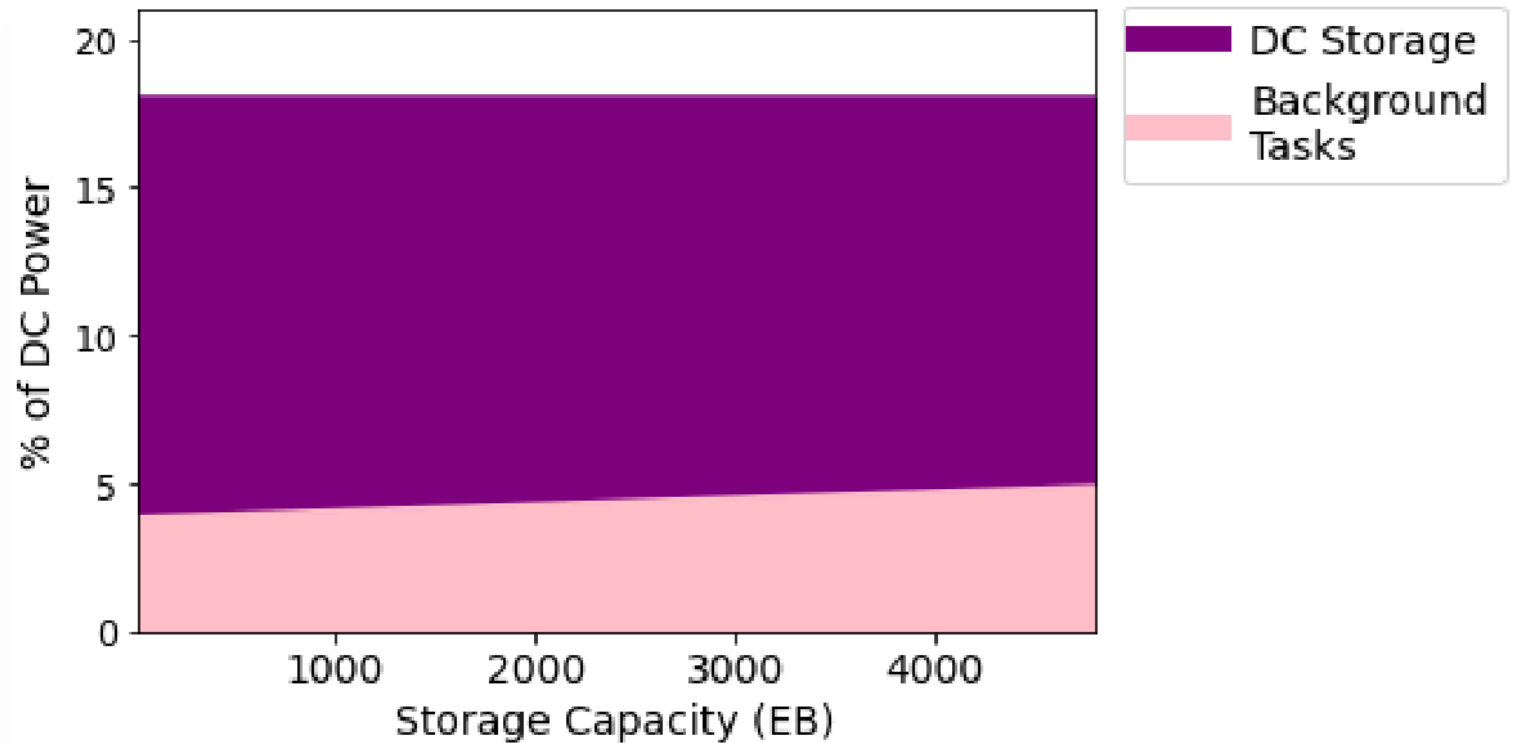
Exercise: space/time trade-off

You want to have access to your model weights over a year. Is it more energy-efficient to store them or to recompute them?

- Payload size: 8 GB
- SSD power draw: 0.5 W/GB (constant)
- Training run duration: 12 hours
- GPU power draw: 1500 W (during training)
- Would [zlib](#) compression with a 2:1 compression ratio change the optimal decision? Assume the power draw for compression is negligible.

Prospective solutions: background tasks

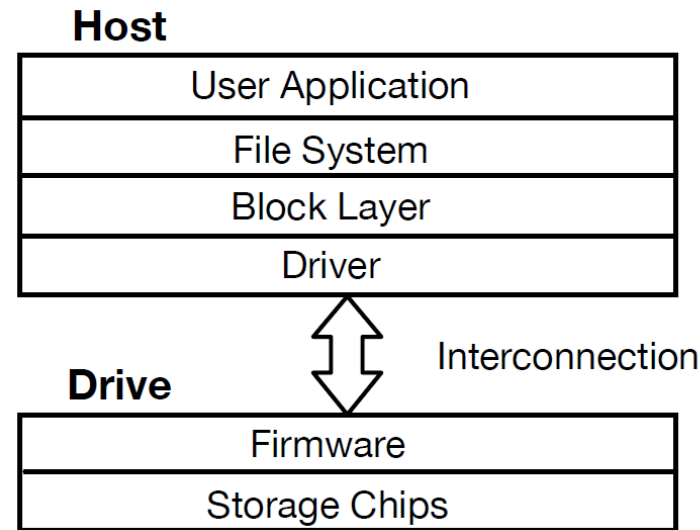
- Background tasks: scrubbing, backup, deduplication etc
 - Critical for reliability and efficiency
 - Temporally flexible
 - 3-5% of [total data center power draw](#)
- Solution: align background tasks with periods of low grid carbon intensity to reduce emissions



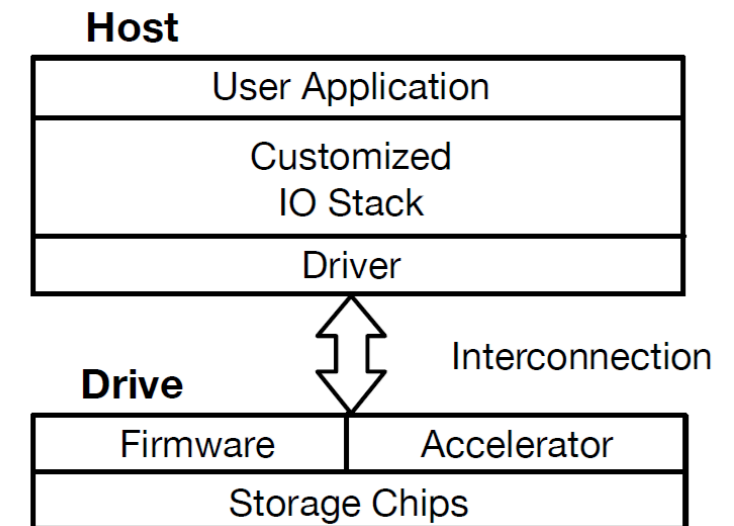
[As cloud storage capacity increases, more reliability and management work is required, increasing power draw](#)

Prospective solutions: in-storage computing

- Recent reductions in SSD latency [bottlenecked](#) by host-side IO stack
- In-storage computing (ISC): integrate [computing substrates](#) within 3D memory array
- Result: [computational storage drives](#) (CSDs) with accelerators significantly improve performance



(a) Classical.

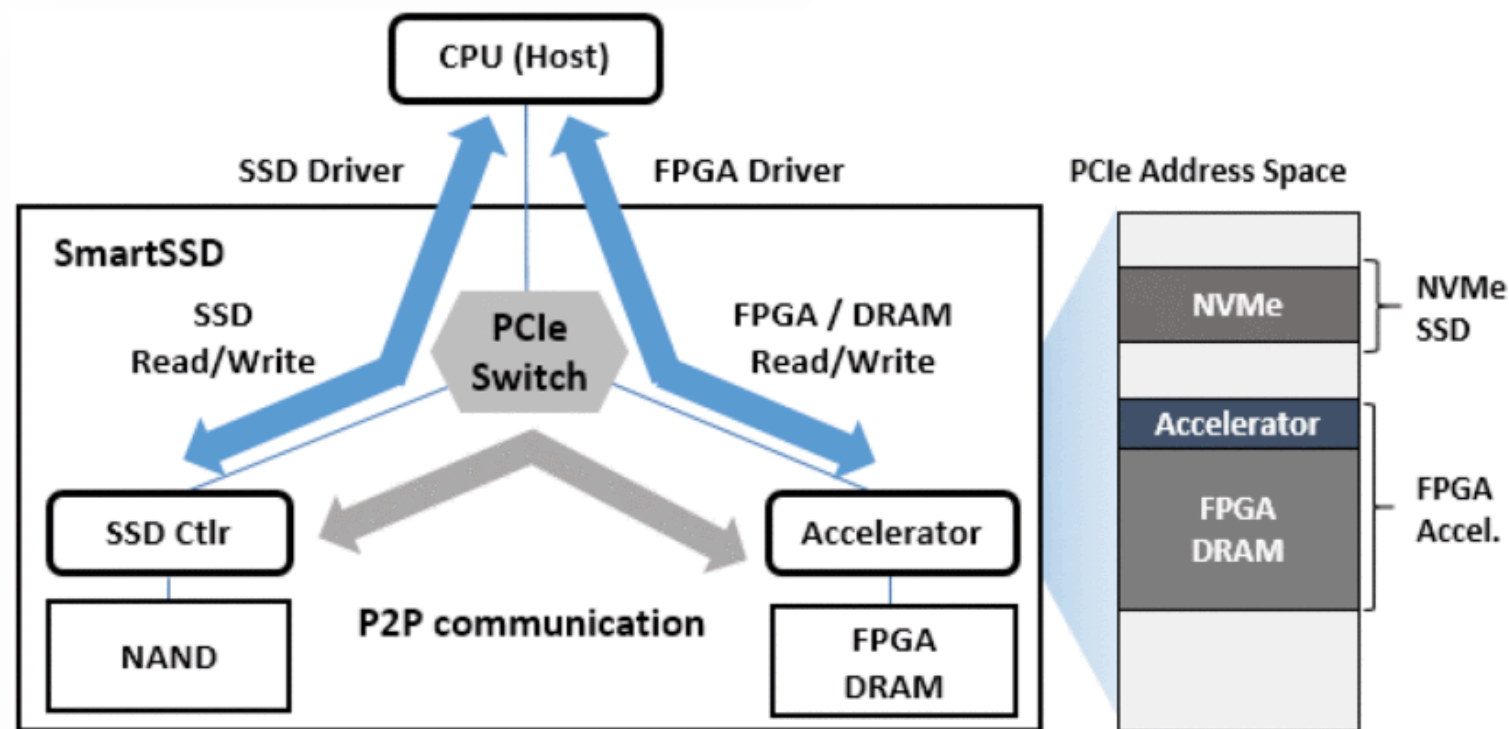


(b) In-storage computing.

[Data processing architecture for classical and in-storage computing drives](#)

Prospective solutions: in-storage computing

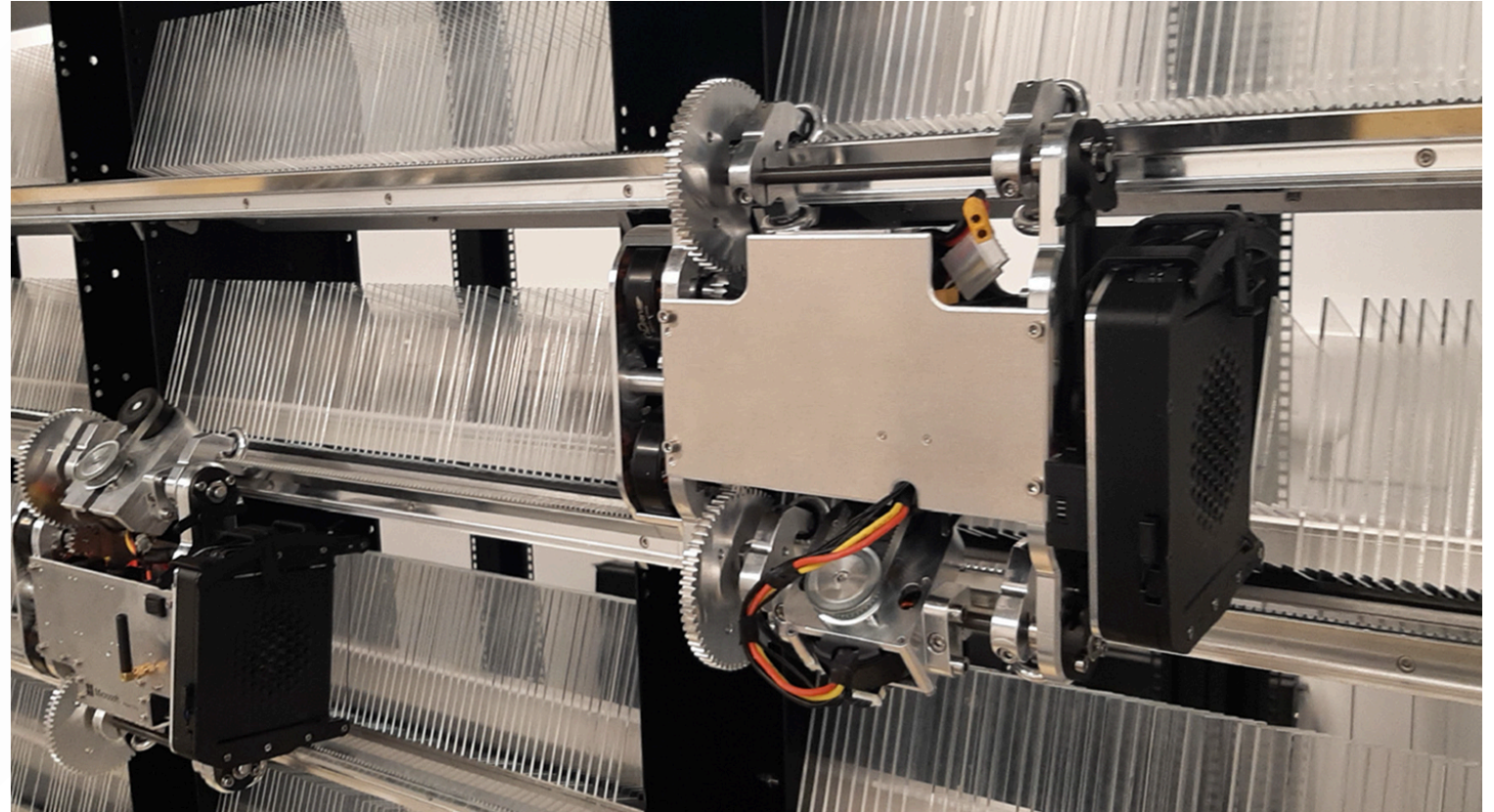
- Up to [62% of computing energy](#) consumed only used to transport data!
- In-storage accelerator handles filtering, reduction → less data movement, [fewer bottlenecks](#), lower power draw
- Lower operational carbon footprint, but embedded carbon grows: [DRAM and compute](#) added to SSD



[Example: an SSD with an internal accelerator can improve performance 3x while reducing power consumption by 50%](#)

Prospective solutions: glass storage

- [Glass storage](#): encoding data into laser pulses to alter physical structure of quartz glass
 - High lifetime (>1000 years), not reversible → archival storage
 - Robotic rail shuttles for writing/reading
 - Expensive ultrashort pulse lasers, otherwise low-cost



[Robotic shuttles in a prototype glass storage archive](#)

Prospective solutions: glass storage

- How does glass storage compare with existing archiving methods?
 - HDD and magnetic tape suffer from [bit rot](#) → energy consumption for background tasks grows over time, replacement needed at end of life
 - Glass storage has no bit rot, theoretically lasts [thousands of years](#) → no replacement needed
 - ... but still in demonstrator phase, no environmental impact assessment available

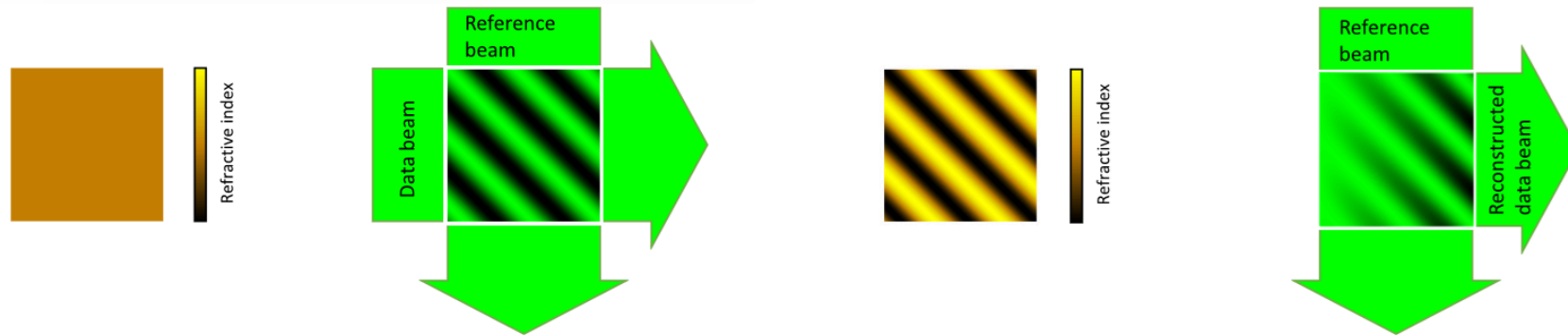
Technology		Tape	Silica
Media Manufacturing	Financial cost	H	L
	Environmental impact	H	L
Media Maintenance	Scrubbing	M	L
	DC Environmentals	H	L
Drive Operations	Read process	M	L
	Write process	M	H
	Processing compute	M	L

L, low; M, medium; H, high.

[Microsoft claims that their Project Silica beats tape in almost all respects](#)

Prospective solutions: holographic storage

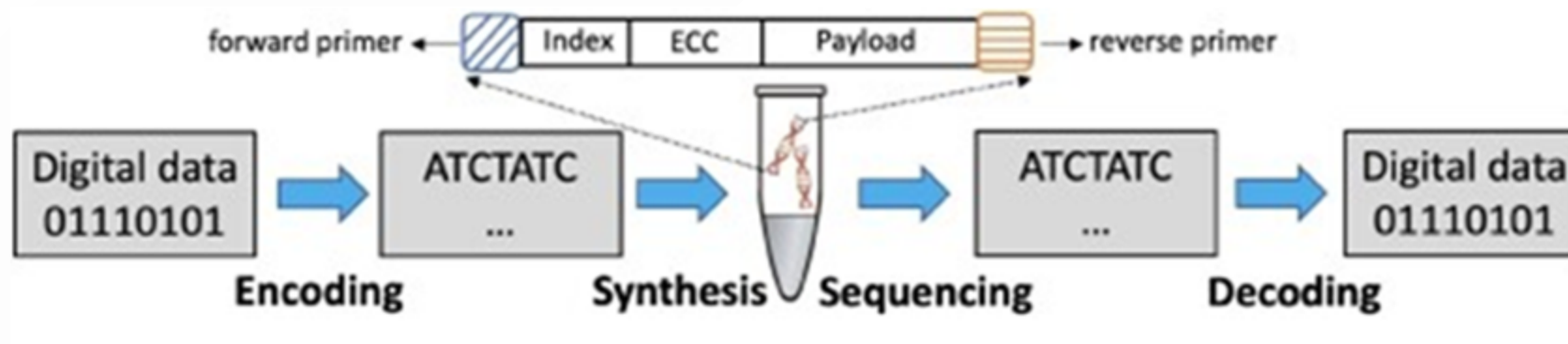
- Holographic storage: recording optical interference (hologram) in photorefractive material
 - [Microsoft research](#): feasible with commercially available laser/camera for writing/reading
 - Only 0.01% of light from reference beam received by camera → very low energy efficiency: 0.65 IOPS/W vs 20-30 IOPS/W for HDDs
 - Significant limitations in currently available materials, fundamental research required



(a) Blank medium has uniform refractive index. (b) Interference pattern creates non-uniformity. (c) Non-uniformity persists as a Bragg grating. (d) Bragg grating diffracts reference beam to recreate data beam.

Prospective solutions: DNA storage

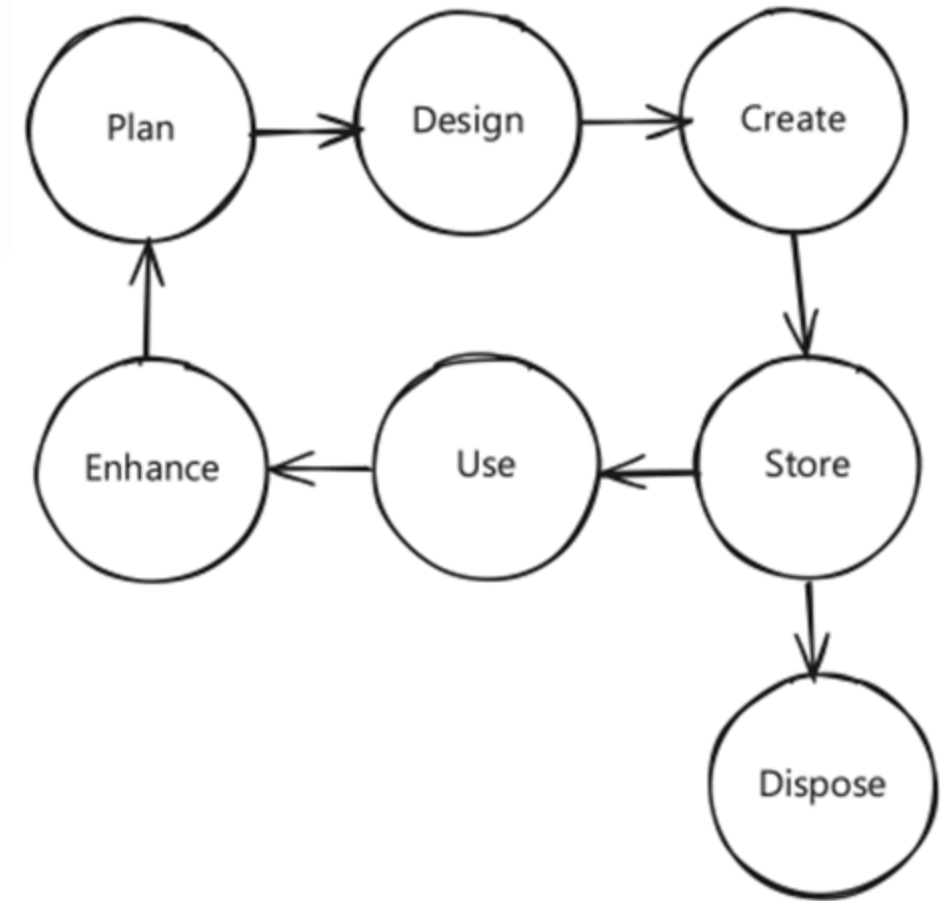
- DNA storage: converting digital data into DNA strands
 - Encoding, synthesis, sequencing, decoding
 - High [theoretical density](#) (455 EB/gram), long-term durability (centuries)
 - ... but subject to biological constraints: hard to store, slow writing/synthesis and reading/sequencing
- Very [limited](#) amounts of data stored in practice (<200 MB), fundamental research required



[Four stages of DNA data storage](#)

Prospective solutions: data strategy

- Software engineering decisions [affect](#) how and what data is stored:
 - Version control: store [deltas](#) rather than versions (e.g. Git)
 - Copying policy: [zero-copy cloning](#) (copy metadata, reference initial files)
 - Deletion policy: mandate temporary retention, then delete



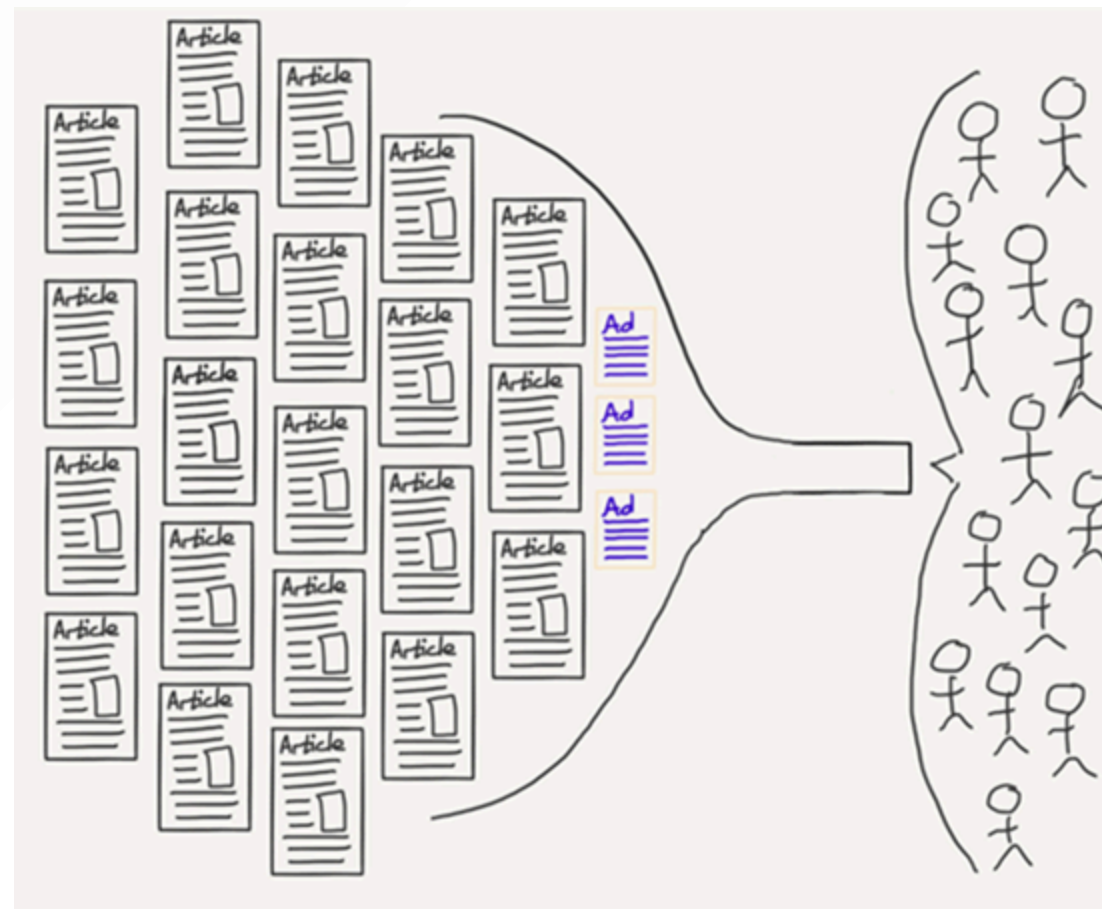
[Seven steps in a sustainable data strategy](#)

Prospective solutions: data sufficiency

What about tackling the problem at the source?

- Higher efficiency (e.g. 3G → 5G) [outpaced](#) by data volumes → rebound effect!
- Increasing number of devices transferring data [without human intervention](#) (e.g. IoT)
- Internet aggregators' [business model](#) generates profit by maximizing user attention and data traffic

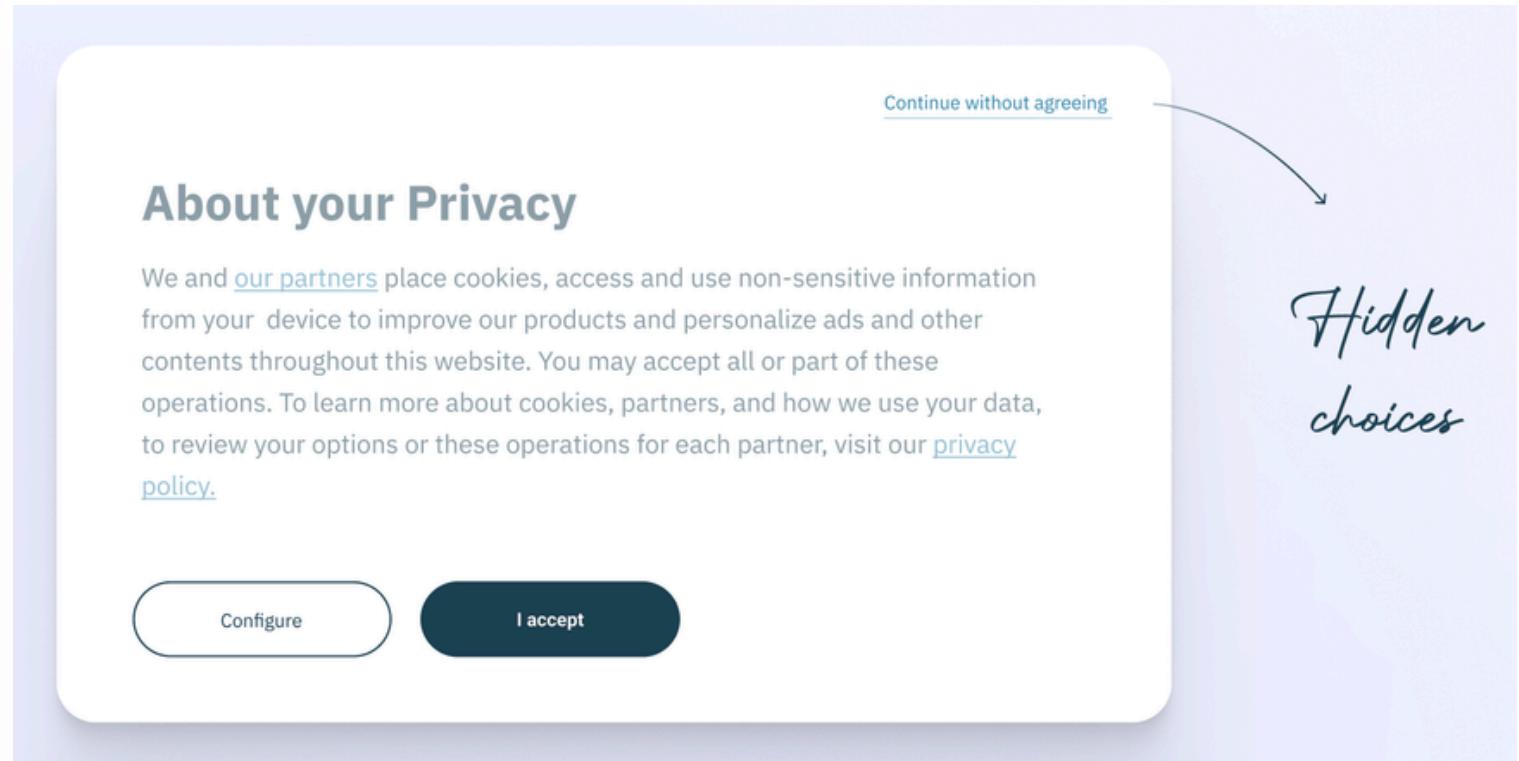
Can these problems be solved by technological innovation?



[Aggregators collect users to leverage access](#)

Prospective solutions: data sufficiency

- [Data sufficiency](#): reducing data creation, traffic, and eventual need for storage
 - Restrict data transfer unnecessary to provide intended functionality (e.g. user behavior)
 - Use open source software/open data standards to prevent redundancy
 - Mandate opt-in rather than opt-out data collection consent



[Dark patterns are sometimes used to manipulate users' data consent decisions](#)

Prospective solutions: data sufficiency

- Nudges: psychological cues to push users towards making a certain decision without outright coercion
 - [Norwegian Consumer Council](#): platforms use nudges for privacy-intrusive defaults
 - [UK Data Act](#): avoid using nudges to manipulate children's online decision-making
- What if nudges were used to help users create and store less data?

Get notifications to free up space

You can set up notifications that alert you when:

- Your storage is almost full
- Apps haven't been used for more than 30 days
- Files take up too much space
- Duplicate files
- Junk files
- Memes

[Nudges and notifications can help users clean up their storage](#)

Recap

- Storage technologies feature both embedded and operational carbon emissions driven by their properties and by the studied use-case
- There is no single optimal storage technology; trade-offs required between embedded carbon footprint, operational energy consumption, and performance
- Improvements in incumbent technologies are showing diminishing returns
- Some significant reductions in environmental footprint can be achieved by leveraging clever architectures and emerging technologies
- Policy decisions about data creation, transfer, and storage required to curb the exponential growth of data

Exam(ple) questions

1. **Question about storage trade-off between HDD and SSD, with annealing for SSDs**
 - a.
 - b.
 - c.
 - d.
2. **Question about storage trade-off between HDD and SSD, with bandwidth limitations**
 - a.
 - b.
 - c.
 - d.

Exam(ple) questions

1. Question about storage trade-off between HDD and SSD, with annealing for SSDs
 - a.
 - b.
 - c.
 - d.
2. Question about storage trade-off between HDD and SSD, with bandwidth limitations
 - a.
 - b.
 - c.
 - d.

Feedback

- We need your feedback to continue improving the course!
- Feedback on Lecture 7.2 slides/notes:



Thank you for your attention!

Good luck on the final presentation and exam!

Questions?